

Investigate Hotel Business using Data Visualization

Hotel Booking & Cancellation Behaviour Analysis

1. Problem Statement

Understanding customer booking behaviour is important for a hotel business because it helps management identify booking patterns, cancellation risks, demand periods, and differences between hotel types. These insights can support decisions related to pricing, staffing, room inventory, promotions, cancellation policies, and revenue planning.

The analysis focuses on three business questions:

Which hotel type do customers book most often: City Hotel or Resort Hotel?

Does the total length of stay affect the booking cancellation rate?

Does lead time, meaning the gap between booking and arrival, affect the cancellation rate?

2. Project Objective

The objective of this project is to clean and analyse hotel booking data and use clear data visualisations to understand hotel-type demand, stay-duration cancellation behaviour, and lead-time cancellation behaviour. The main deliverable is a business-focused analysis that converts these findings into practical recommendations for hotel management.

3. Data Overview

The dataset contains 119,390 booking records and 29 columns, covering hotel bookings from 2017 to 2019. The analysis uses fields related to hotel type, booking status, arrival date, stay duration, lead time, guest counts, meal, agent/company information, and average daily rate (ADR).

Key columns used for this analysis include hotel, is_canceled, lead_time, arrival_date_year, arrival_date_month, stays_in_weekend_nights, stays_in_week_nights, adults, children, babies, adr, meal, agent, company, and country/city-related fields.

4. Data Assessment

4.1 Missing Values

Missing values were assessed before analysis. The main affected fields were children, city, agent, and company. The children field had only 4 missing records and was filled with 0 because a missing value in this context can reasonably be treated as no children recorded. Missing agent and company values were retained as missing because they indicate that no agent/company was associated with the booking; assigning an artificial agent or company would create misleading information. City-related missing values were also retained/treated as unknown rather than inventing a location.

4.2 Duplicate Rows

The dataset contained 33,261 duplicate rows. Duplicate records were removed so that repeated observations would not artificially increase booking and cancellation counts.

4.3 Inconsistent or Unclear Values

The meal column contained an 'Undefined' category. This value was treated as an 'Undefined/Unknown' meal category rather than being interpreted as a specific meal type. This preserves the original information without assigning a meal plan that was not recorded.

4.4 Anomalies

One record contained a negative ADR, which is not a valid average daily rate, so it was excluded from the analytical dataset. Records with zero total guests were also excluded because they do not represent a meaningful guest booking for behavioural analysis.

After duplicate removal and anomaly handling, the cleaned analytical dataset contained 85,963 records.

5. Monthly Booking Analysis by Hotel Type

5.1 Booking Share: City Hotel vs Resort Hotel

City Hotel was booked more often than Resort Hotel. In the cleaned dataset, City Hotel represented approximately 60.98% of bookings, while Resort Hotel represented approximately 39.02%. Therefore, City Hotel had the larger share of customer bookings.

This indicates stronger overall booking demand for the City Hotel, although the Resort Hotel still represents a substantial portion of total demand.

5.2 Monthly Booking Trend

Monthly booking volume showed clear variation across the year. Considering total bookings after cleaning, October was the busiest month with 11,135 bookings, while March was the quietest with 4,624 bookings.

The monthly pattern indicates that demand is seasonal. Higher booking volumes during stronger travel periods can be associated with holidays, vacation planning, favourable travel periods, and business travel. These are possible business explanations for the observed pattern rather than effects directly proven by the dataset.

The monthly trend can help hotels plan room inventory, staffing, promotions, and pricing. During high-demand months, hotels can increase prices where demand supports it, prepare sufficient staffing, and protect room availability. During lower-demand months, targeted promotions and packages can be used to stimulate demand.

6. Impact of Stay Duration on Cancellation Rate

6.1 Cancellation Rate by Hotel Type

City Hotel had the higher overall cancellation rate. In the cleaned analytical data, approximately 30.22% of City Hotel bookings were cancelled compared with approximately 23.49% of Resort Hotel bookings.

Therefore, cancellation risk was higher for City Hotel bookings overall.

6.2 Stay Duration and Cancellation Trend

The relationship between stay duration and cancellation was not perfectly linear for both hotel types. For City Hotel, cancellation generally became higher as stay duration increased, particularly for longer stays. For example, the cancellation rate increased from approximately 22.37% for one-night stays to around 70.45% for 14-night stays among the analysed duration groups.

For Resort Hotel, cancellation also increased through the shorter-to-medium stay range, but the pattern became less consistent at longer durations. For example, the rate was approximately 12.53% for

one-night stays, around 30.54% for five-night stays, and varied at longer durations.

Overall, the relationship is more clearly positive for City Hotel, while Resort Hotel shows a weaker and less consistent relationship.

6.3 Possible Reasons

Longer stays may involve plans made further in advance, giving customers more opportunity to change or cancel their plans.

Longer trips may have higher total booking costs, making customers more sensitive to changes in travel plans or price.

Resort and City Hotel customers can have different travel purposes, which may produce different cancellation behaviour.

7. Impact of Lead Time on Cancellation Rate

7.1 Definition and Relationship

Lead time is the number of days between the date a booking is made and the guest's scheduled arrival date. It measures how far in advance a customer makes the reservation.

The analysis shows a clear relationship between longer lead times and higher cancellation risk, especially for City Hotel. For City Hotel, cancellation increased from approximately 10.54% for bookings made 0–7 days before arrival to 45.18% for bookings made 181–365 days in advance and 52.82% for bookings made 366 or more days in advance.

For Resort Hotel, cancellation increased from approximately 5.95% for 0–7 day lead time to 34.84% for 181–365 days. The 366+ group was lower at approximately 21.39%, but it contains relatively few bookings, so this result should be interpreted cautiously.

7.2 Comparison of Hotel Types

City Hotel shows the sharper and more consistent rise in cancellation risk as lead time increases. Resort Hotel also shows an increasing pattern through most lead-time groups, but the pattern is less steep and less stable at the longest lead time.

A possible explanation is that far-ahead bookings provide more time for customers to change plans. Differences in customer purpose, booking channels, pricing, and cancellation terms may also contribute to the different patterns between hotel types.

8. Summary and Recommendations

8.1 Key Findings

City Hotel is the more frequently booked hotel type, accounting for approximately 60.98% of cleaned bookings.

October was the busiest month and March was the quietest month in the cleaned data.

City Hotel had the higher overall cancellation rate: approximately 30.22% versus 23.49% for Resort Hotel.

Longer stays are associated with higher cancellation risk more clearly for City Hotel, while Resort Hotel shows a less consistent relationship.

Cancellation risk generally increases as lead time becomes longer, with the strongest rise observed for City Hotel.

8.2 Recommendations Based on Hotel Type and Seasonality

Use peak-month demand to optimise room pricing, staffing, and inventory, particularly around the higher-demand months such as October.

Use targeted promotions, packages, or value-based offers during quieter periods such as March to improve occupancy.

Since City Hotel has the larger booking share, maintain strong inventory and service capacity while also developing targeted promotions to increase Resort Hotel demand.

8.3 Stay Duration and Cancellation Policies

For longer City Hotel stays, the hotel can consider stricter cancellation terms, deposits, or partially non-refundable rate options. Flexible rates can still be offered at a higher price so customers can choose flexibility while reducing the hotel's exposure to cancellation losses.

8.4 Lead-Time Actions for Far-Ahead Bookings

Because cancellation risk increases substantially for far-ahead City Hotel bookings, the hotel should consider confirmation reminders, deposit requirements, flexible rescheduling options, and differentiated pricing for advance bookings. These actions can reduce uncertainty while preserving the ability to attract customers who book early.

8.5 Single Biggest Recommendation

The biggest priority should be reducing cancellations from far-ahead City Hotel bookings. The lead-time analysis shows a strong increase in City Hotel cancellation rates, reaching approximately 45.18% for bookings made 181–365 days in advance and 52.82% for bookings made 366+ days in advance. A combination of advance-booking deposits, reminder messages, and controlled cancellation/rescheduling terms could therefore have a meaningful impact on protecting expected room revenue.